POST 4/50

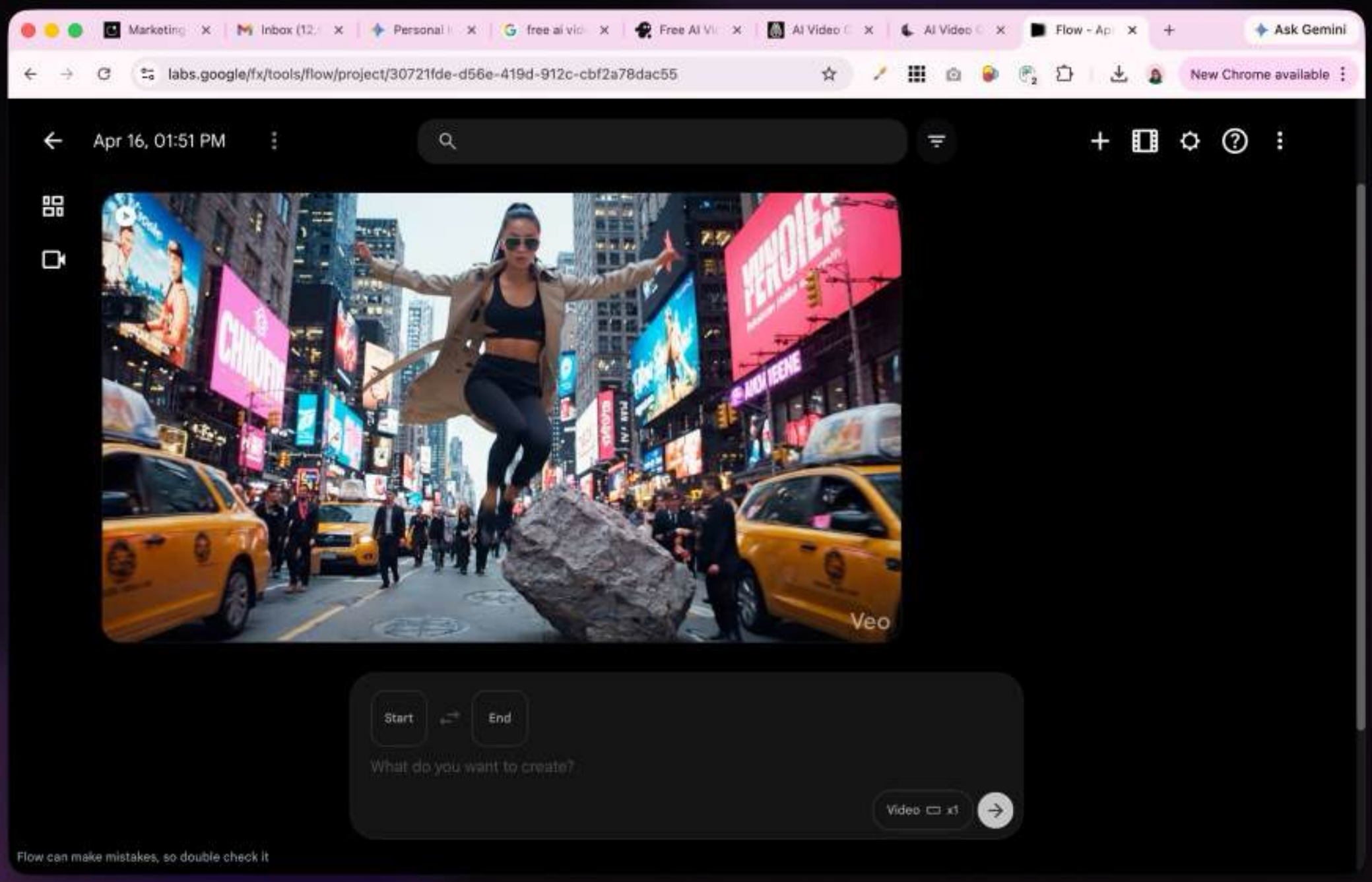
Speculative Redesign of ‘Google Flow’, an AI-Native Video Editing Tool

ZEAL SHETH
AI-FIRST PRODUCT DESIGNER + STRATEGIST

- The creative landscape is evolving at lightning speed, and our tools need to keep up.
- I recently took a hands-on approach with frontier AI models like Veo, Nano banana and Generative AI tools to explore critical gap between "Tools We Have" and "Tools We Need" in an AI-powered world.
- In this post, I deep-dive into speculative redesign of ‘Google Flow’ - An AI-native video editing tool to empower creatives from filmmakers and marketers to animators and VFX artists.

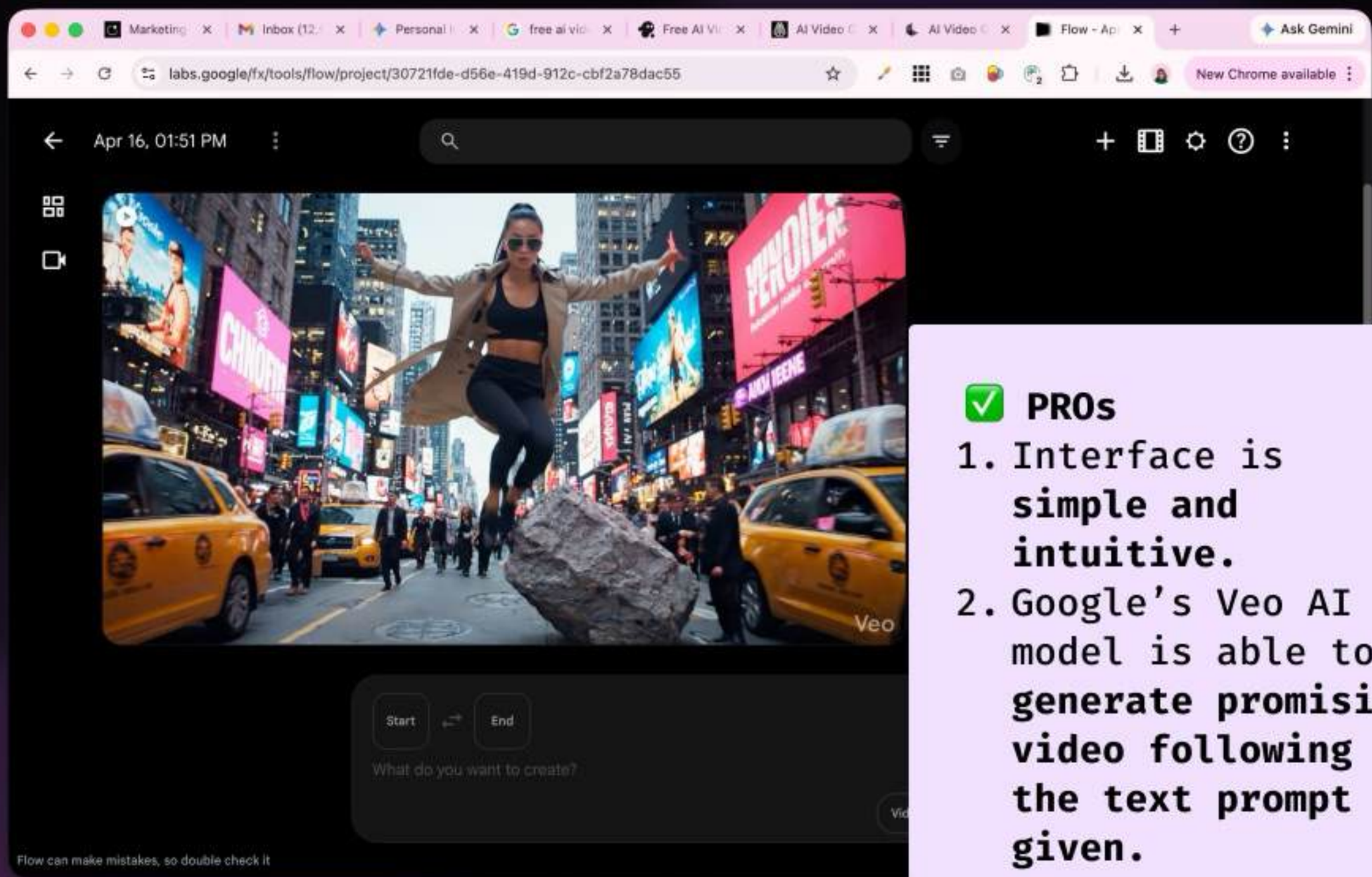
Current State of Google Flow

- I conducted a UX audit to understand Flow's strengths and identify areas ripe for AI-driven transformation.
- To perform the UX Audit: I roleplayed as a Marketing Ad Campaign Designer for Sunglasses company. I created a 30sec Video Ad for Sunglasses with a rough script.



Current State of Google Flow

- I conducted a UX audit to understand Flow's strengths and identify areas ripe for AI-driven transformation.
  - To perform the UX Audit: I role as a Marketing Ad Campaign Designer for Sunglasses company. I created a 30sec Video Ad for Sunglasses with a rough script.

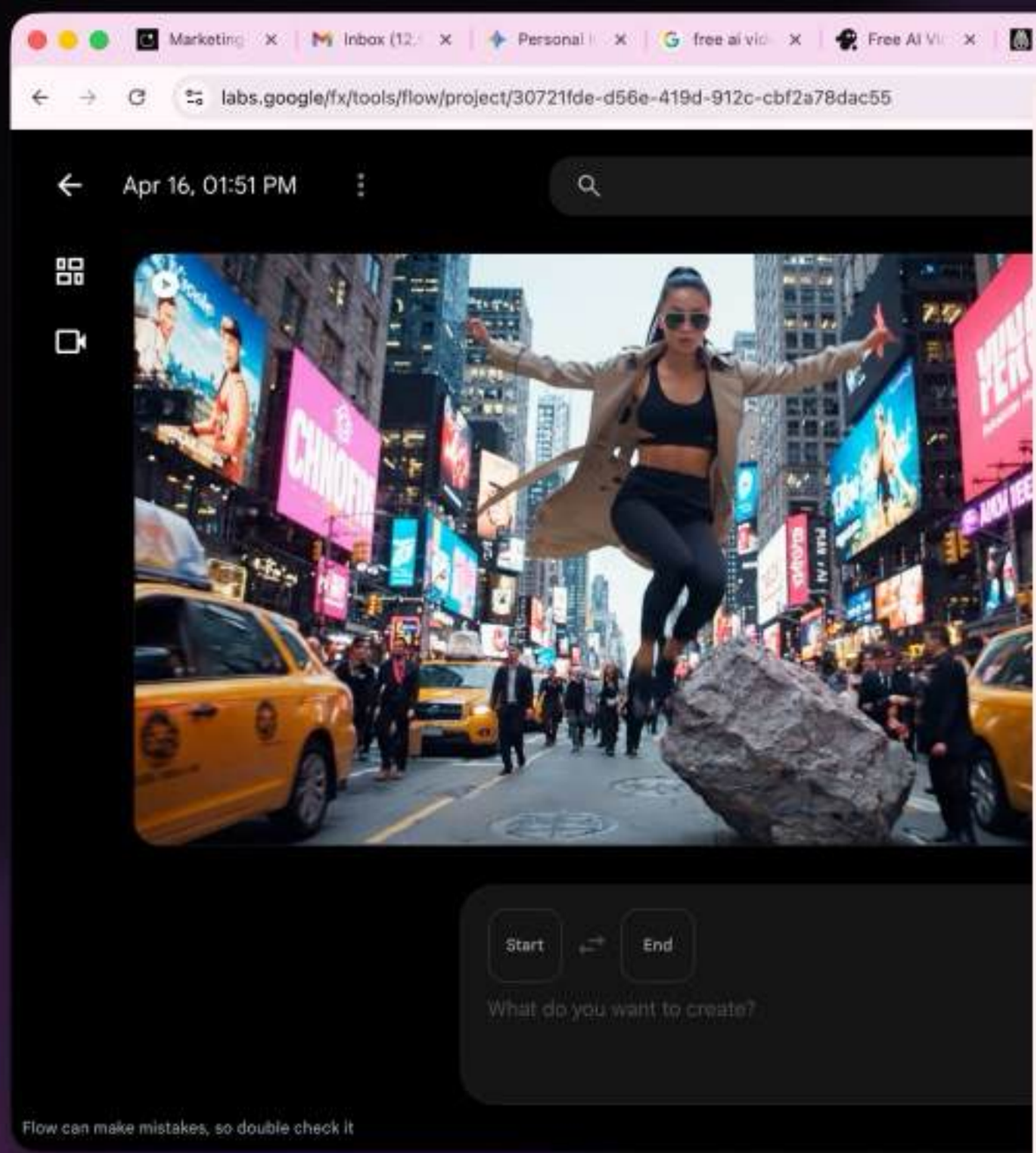


✅ **PROs**

1. Interface is **simple and intuitive.**
2. Google's Veo AI model is able to **generate promising video following the text prompt given.**

Current State of Google Flow

- I conducted a UX audit to understand Flow's strengths and identify areas ripe for AI-driven transformation.
  - To perform the UX Audit: I role as a Marketing Ad Campaign Designer for Sunglasses company. I created a 30sec Video Ad for Sunglasses with a rough script.



❌ **CONs**

1. Making **minor tweaks in the scene is difficult** with just providing the text chat instruction.
2. Every time some changes are made with prompting, **there is no way to lock the parts of the video which you don't want to be changed.**
3. It is a **heavy AI token usage process** even in case of minor tweaks making it restrictive for creatives.

‘How might we enhance generative AI video editing capabilities for creatives with Google Flow? ’

Background Research

- I took a Top-down and Bottom-up approach for this redesign.
- For the Top-down approach, I did extensive research on real user needs for creatives to understand where this tool can generate value. Then, with the Bottom-up approach I identified the different functions that can be achieved with the current AI models.
- Mapping the two to find intersection can provide us with the features that are actually viable and can create real user impact.

Feature Mapping with Technical Viability

Feature Mapping with Technical Viability

Creative Needs

- Object Recognition and Tweaking (for animation/motion/visual appearance)
- Chat with Scene (for giving instructions on movement, emotion, appearance/disappearance)
- Adding Graphic Elements (for Text, Illustrations, Shapes, Interactive Elements, Particle Effects insertion)
- Video Stabilization
- Lighting and Color Adjustments
- Object Consistency (for Character/Environment)
- Camera Shot Control

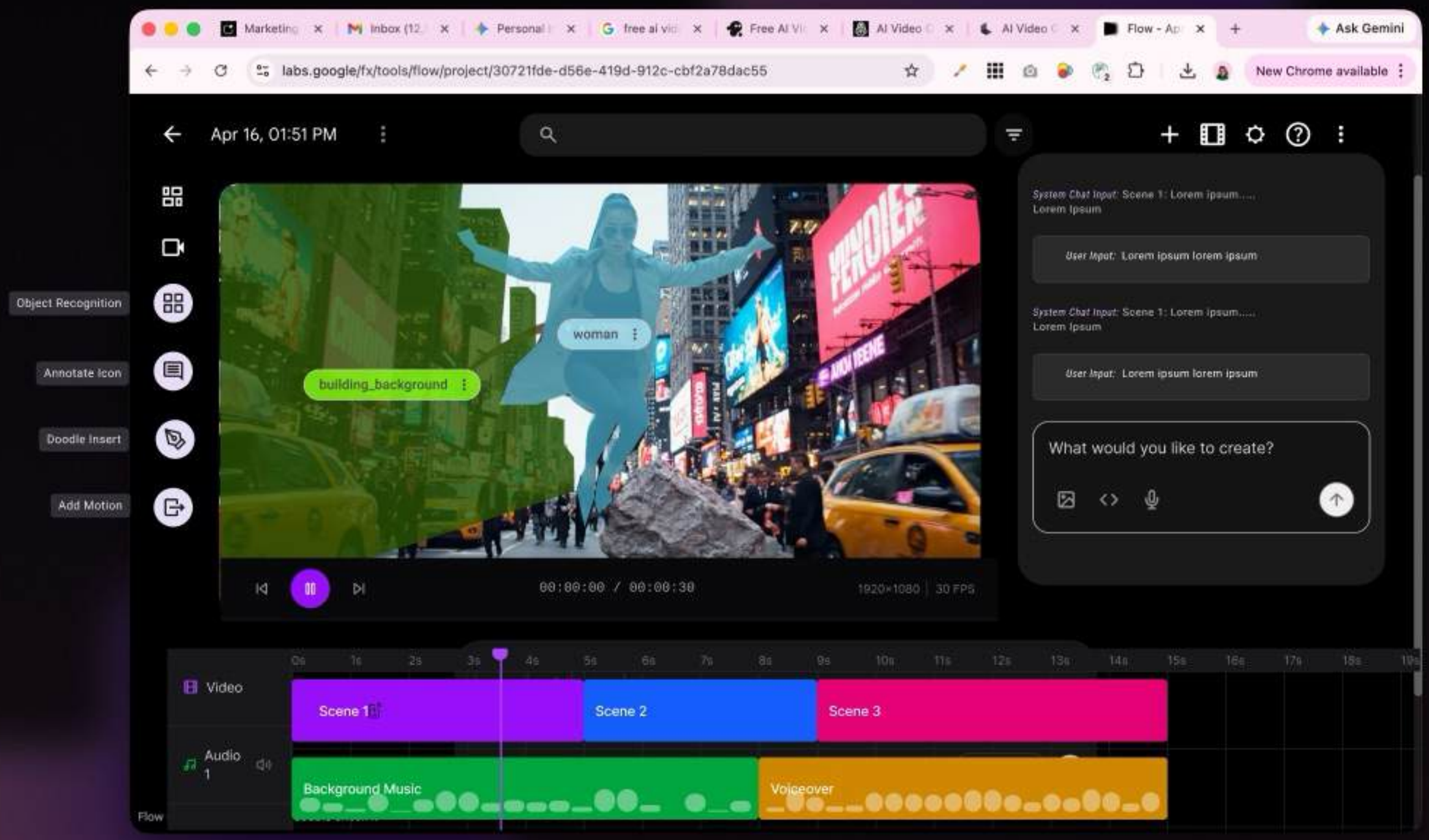
AI Models Capabilities

- Image Segmentation
- Image Classification + LLMs (for natural language chat)
- Generative AI Image Creation with Nanobanana, ImageFX (for Area Selector, Doodling, Reference Images, Image Planner/Agent.)
- 3D Object AI Generation from Text eg. DreamFusion
- Space-Time Diffusion Models eg. Lumiere
- AI Video Script Generator eg. YouTube Creator Studio
- AI Editing/Stabilization Suggestion eg. YouTube Creator Studio
- AI Music Generator eg. Lyria Model, MusicFX
- Basic AI Video Hygiene Functions eg. Google Meet (for Noise Cancellation, Auto-Caption, Auto-Blurring)
- AI Lighting and Color Correction eg. Google Photos
- Animating Still Images eg. Google Photos
- Object (2D/3D) Motion Animation with Genie Model, Gemini Robotics, Google 360 (for motion design/VFX)
- Camera Shot Suggestions with Genie Model (for visual rendering, shot direction)

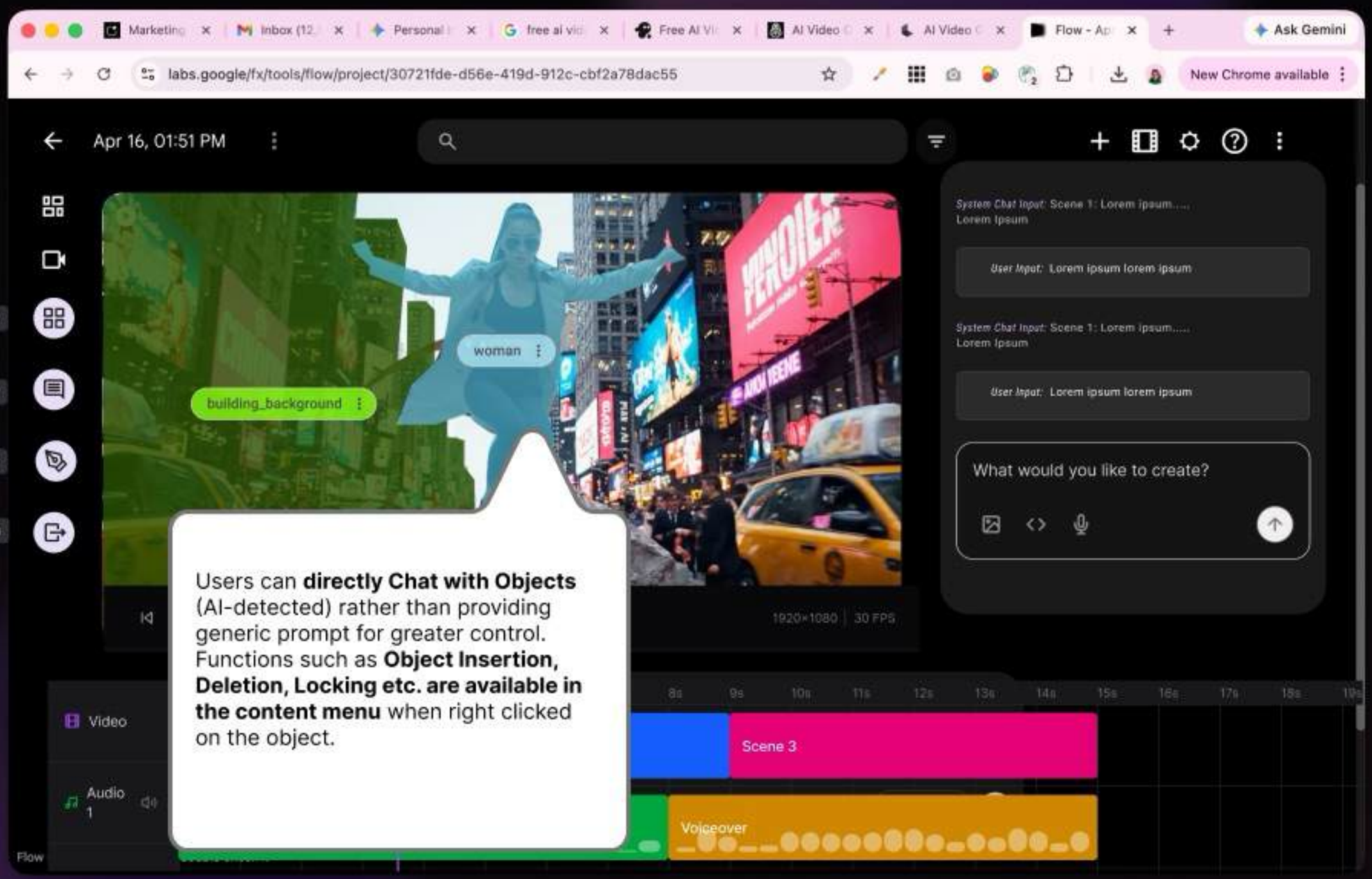
Steps for AI-Native Product UX Design

1. **Multimodal Interaction Design** - To make interaction intuitive, blurring the lines between real and virtual world.
2. **Adaptive AI Design & Error Handling** - To fine-tuning outcome and avoid AI Slop.
3. **Sustainable AI Usage Design** - For sustainable and efficient AI token usage.
4. **AI Guardrails Design** - To avoid bad actors behavior for privacy, security and safety.
5. **Design for Credibility** - To avoid content plagiarism, deepfakes, harmful content for the society.

Redesign Highlights – Multimodal Interaction

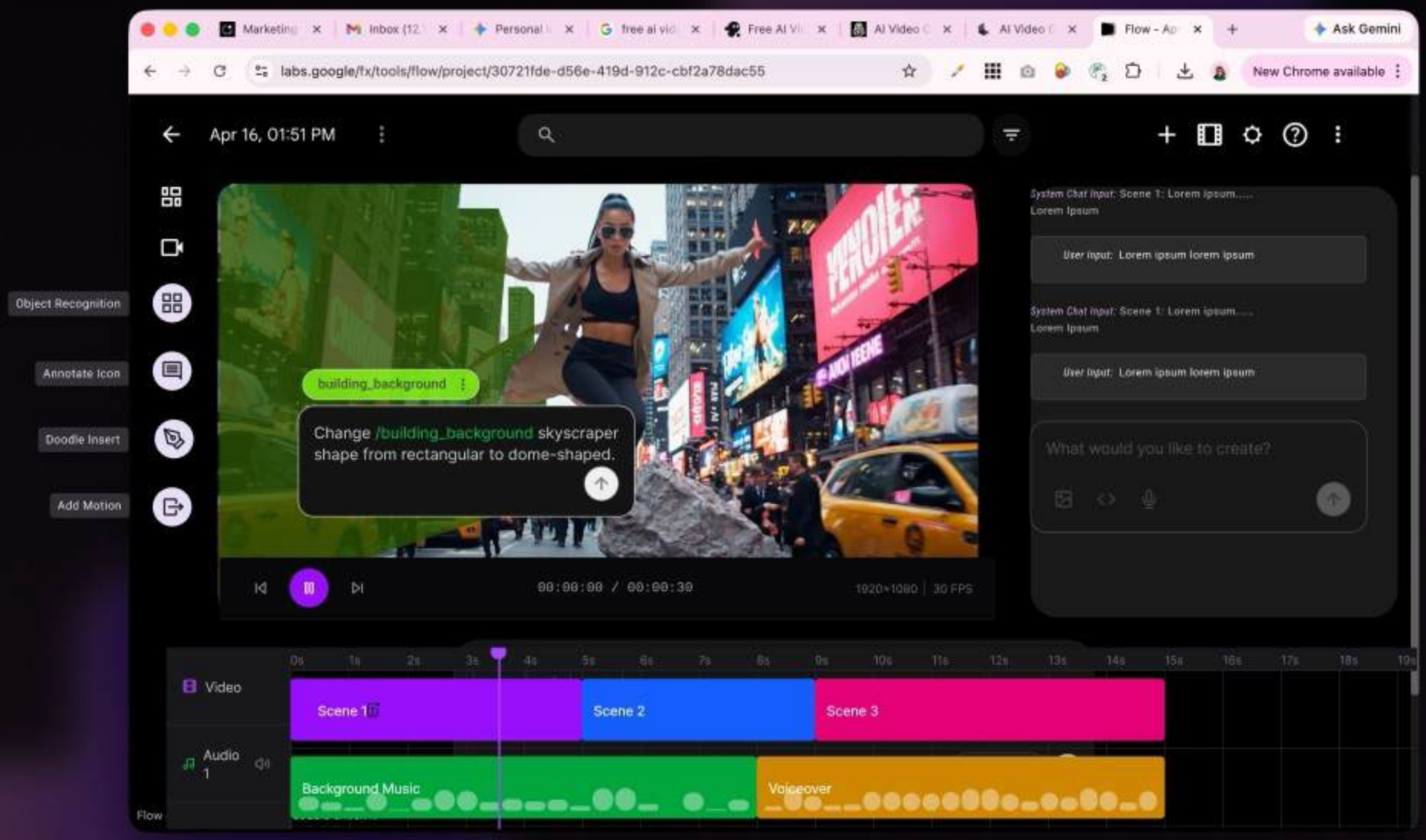


Redesign Highlights – Multimodal Interaction

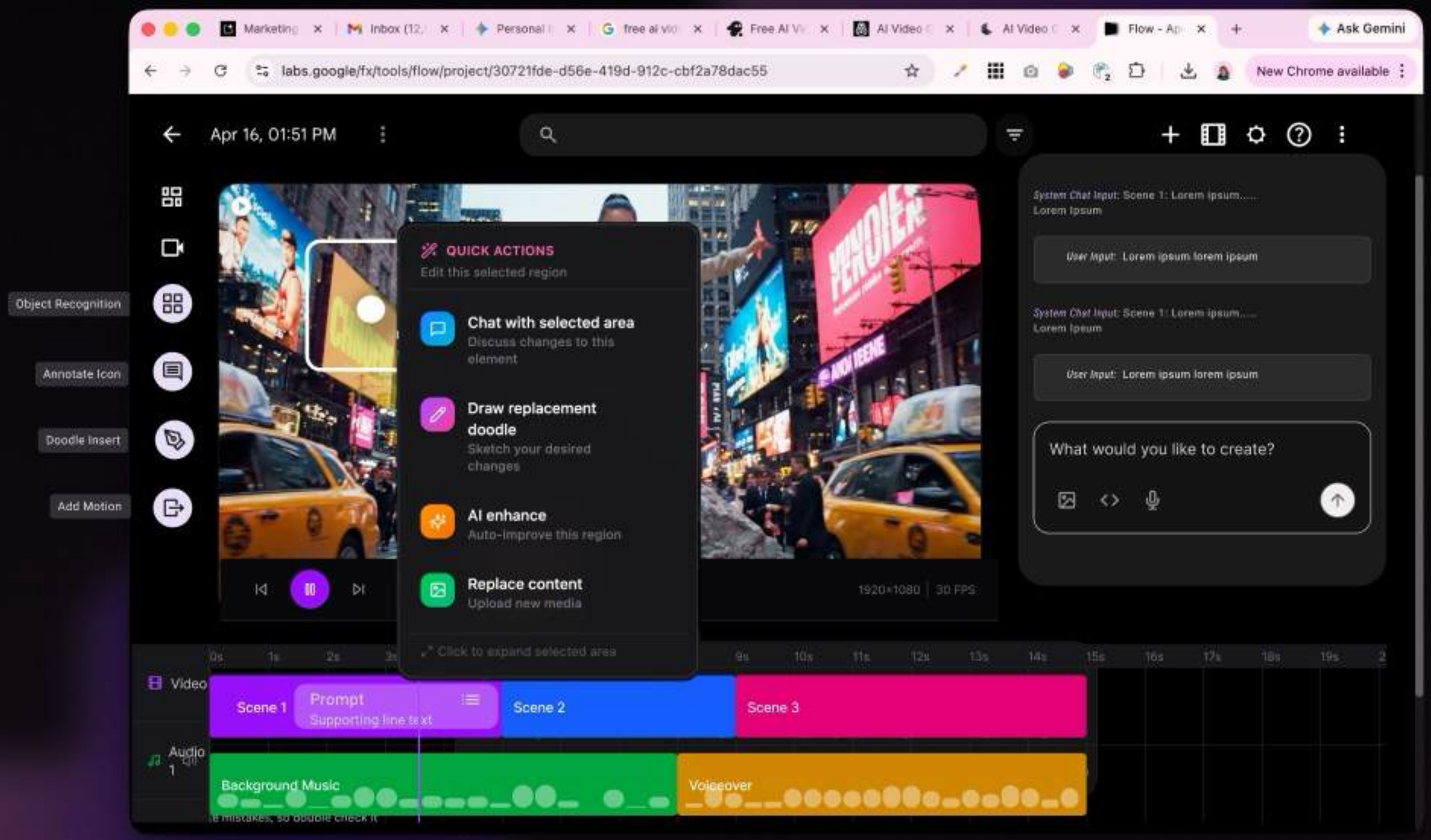


Users can **directly Chat with Objects** (AI-detected) rather than providing generic prompt for greater control. Functions such as **Object Insertion, Deletion, Locking etc. are available in the content menu** when right clicked on the object.

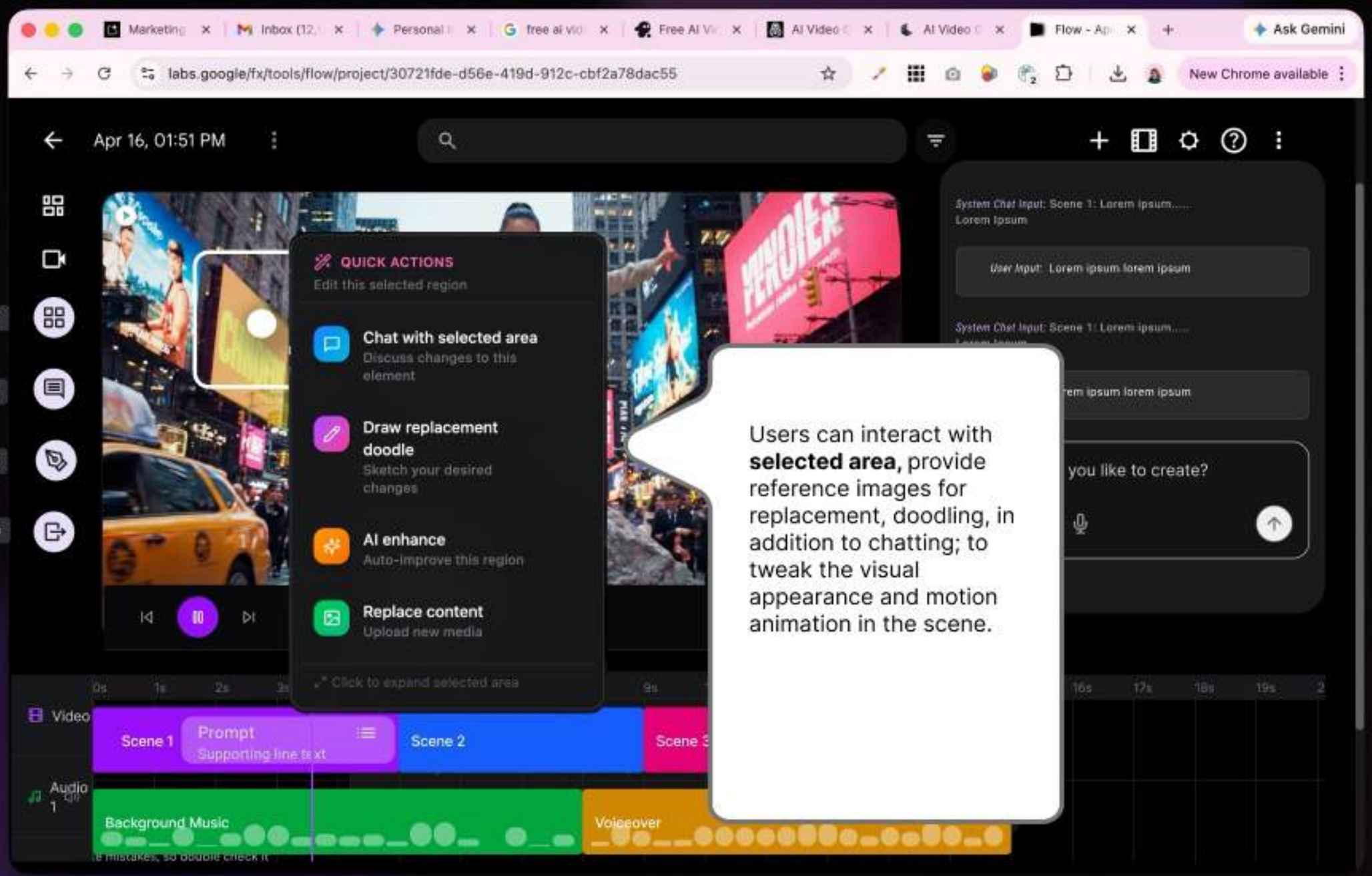
Redesign Highlights – Multimodal Interaction



Redesign Highlights – Multimodal Interaction

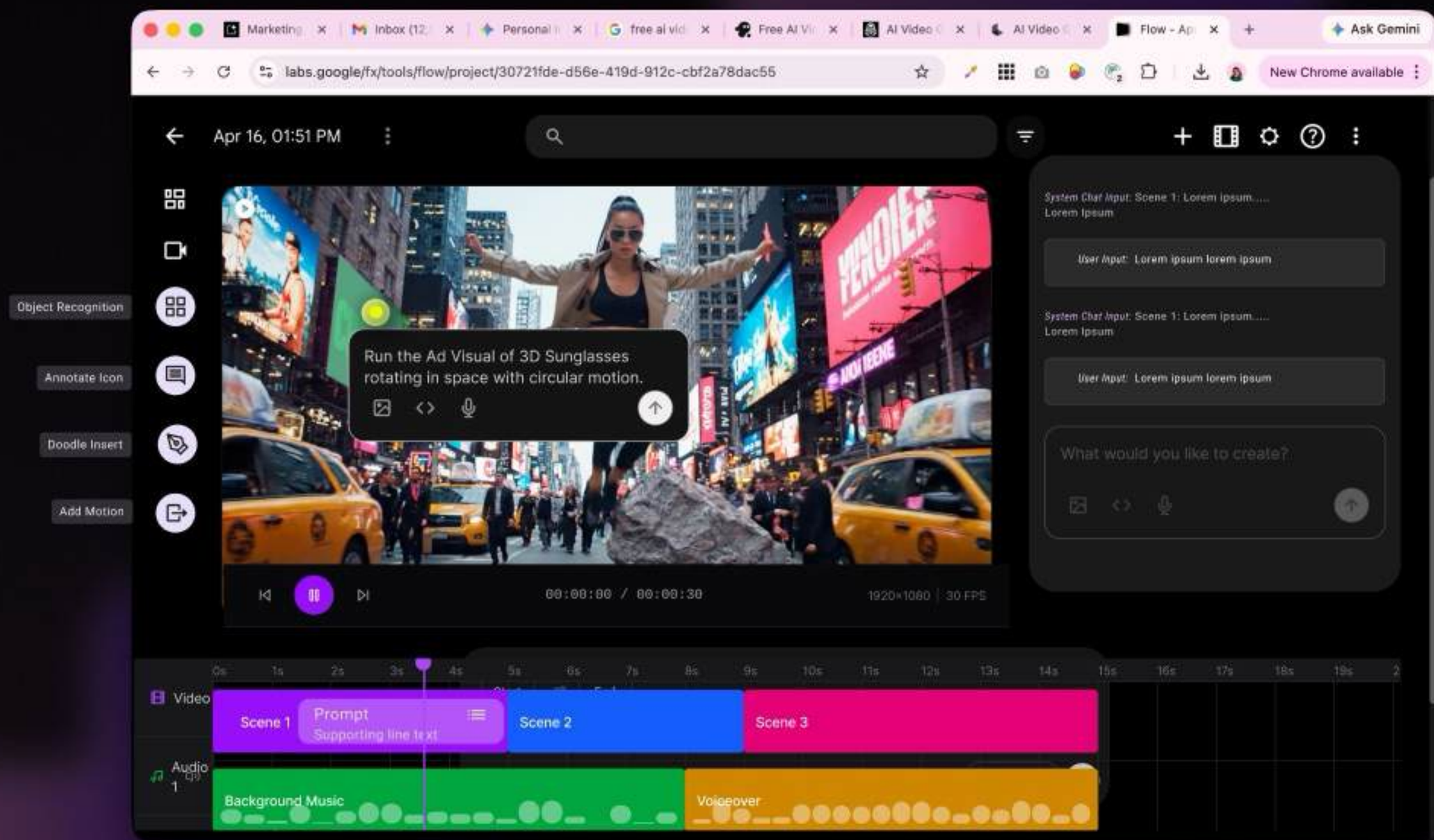


Redesign Highlights – Multimodal Interaction

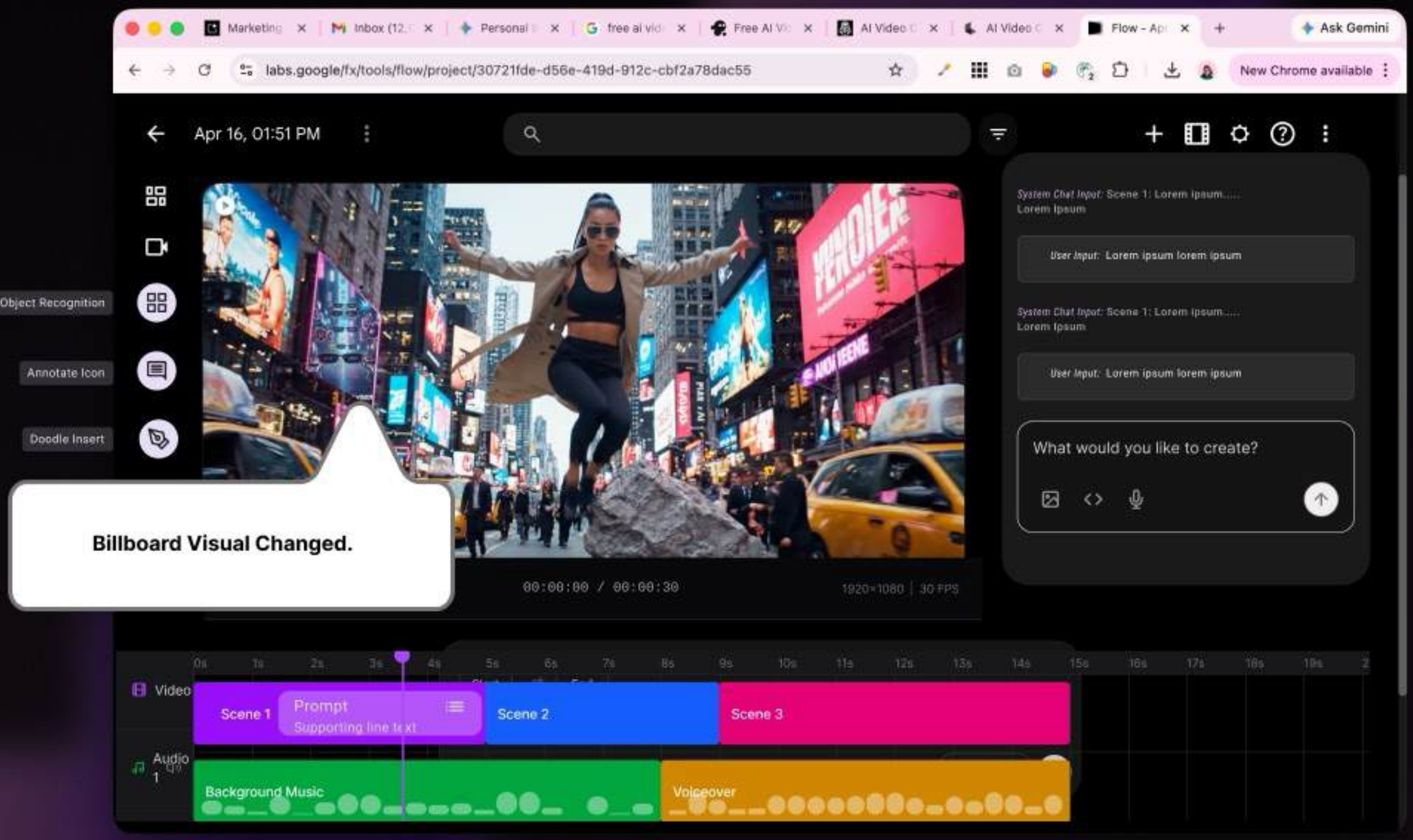


Users can interact with **selected area**, provide reference images for replacement, doodling, in addition to chatting; to tweak the visual appearance and motion animation in the scene.

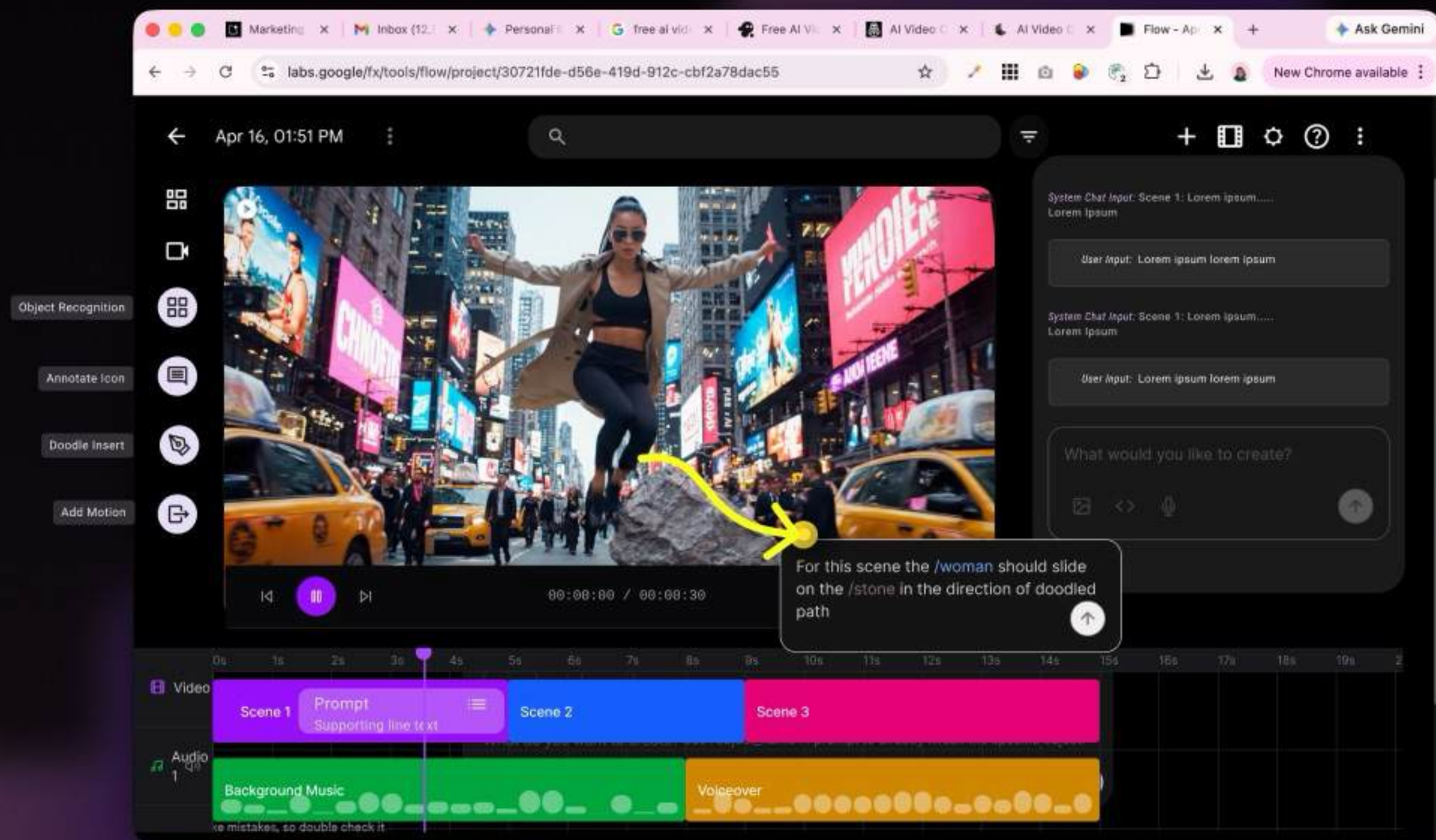
Redesign Highlights – Multimodal Interaction



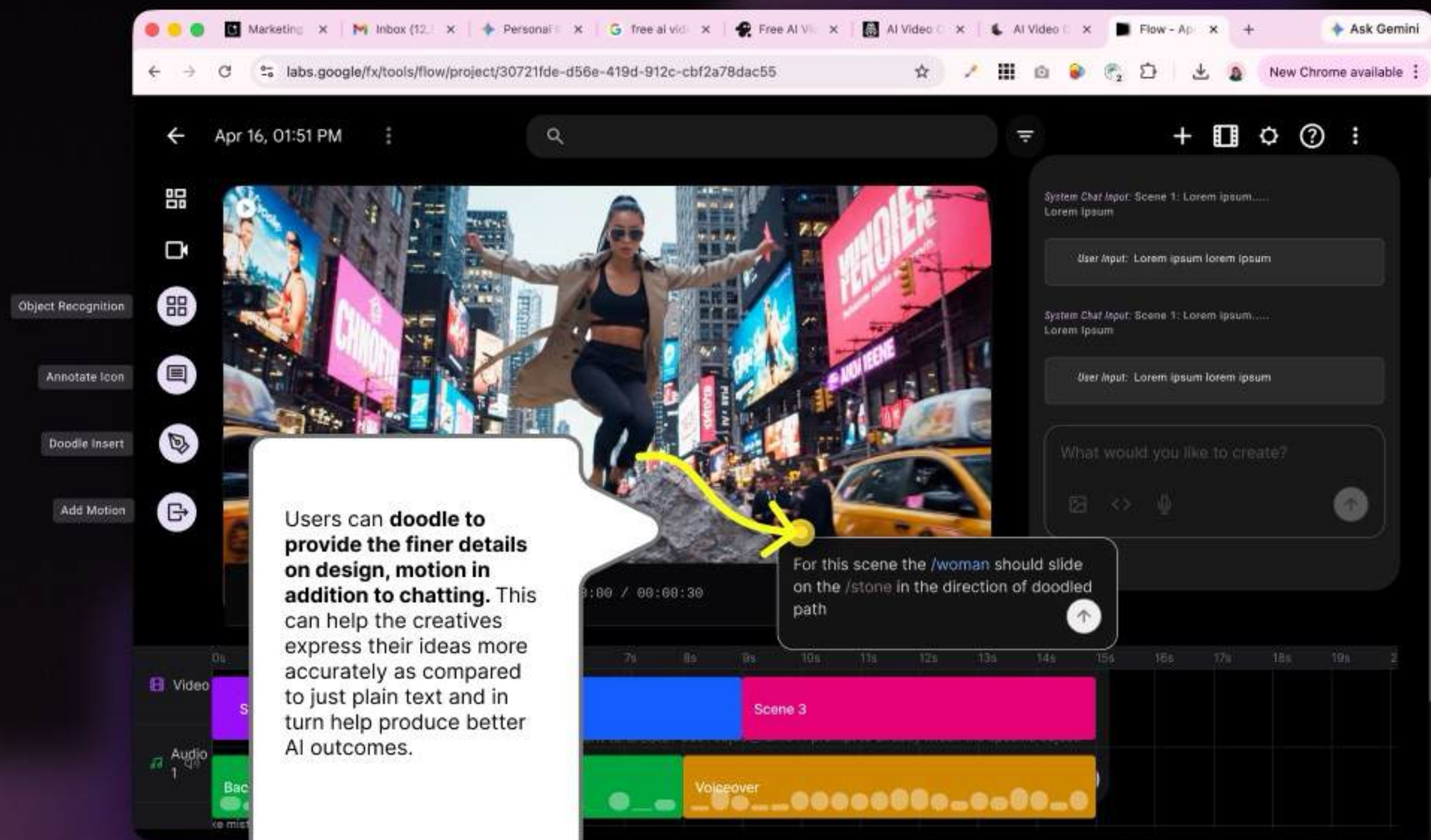
Redesign Highlights – Multimodal Interaction



Redesign Highlights – Multimodal Interaction

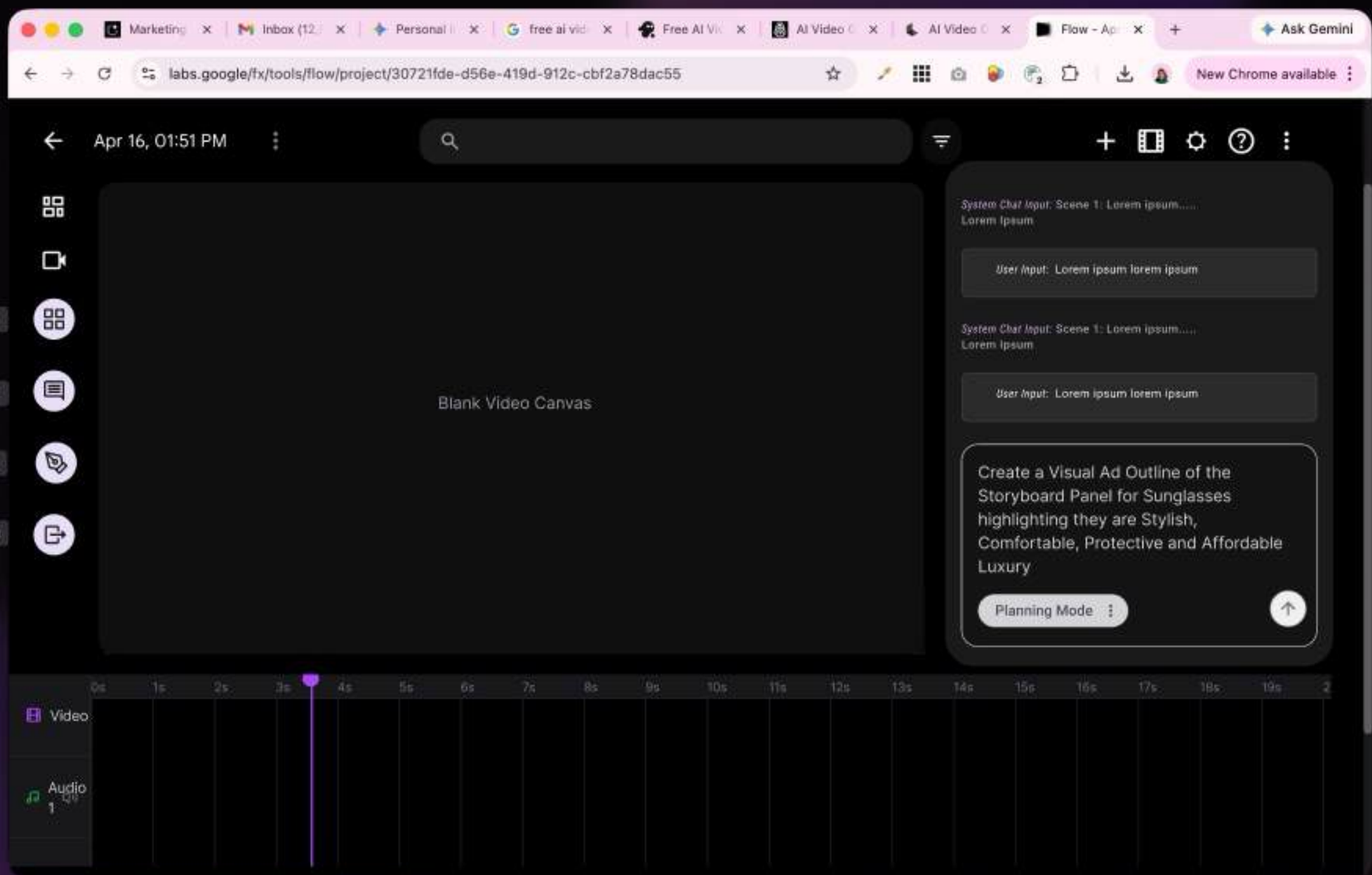


Redesign Highlights – Multimodal Interaction

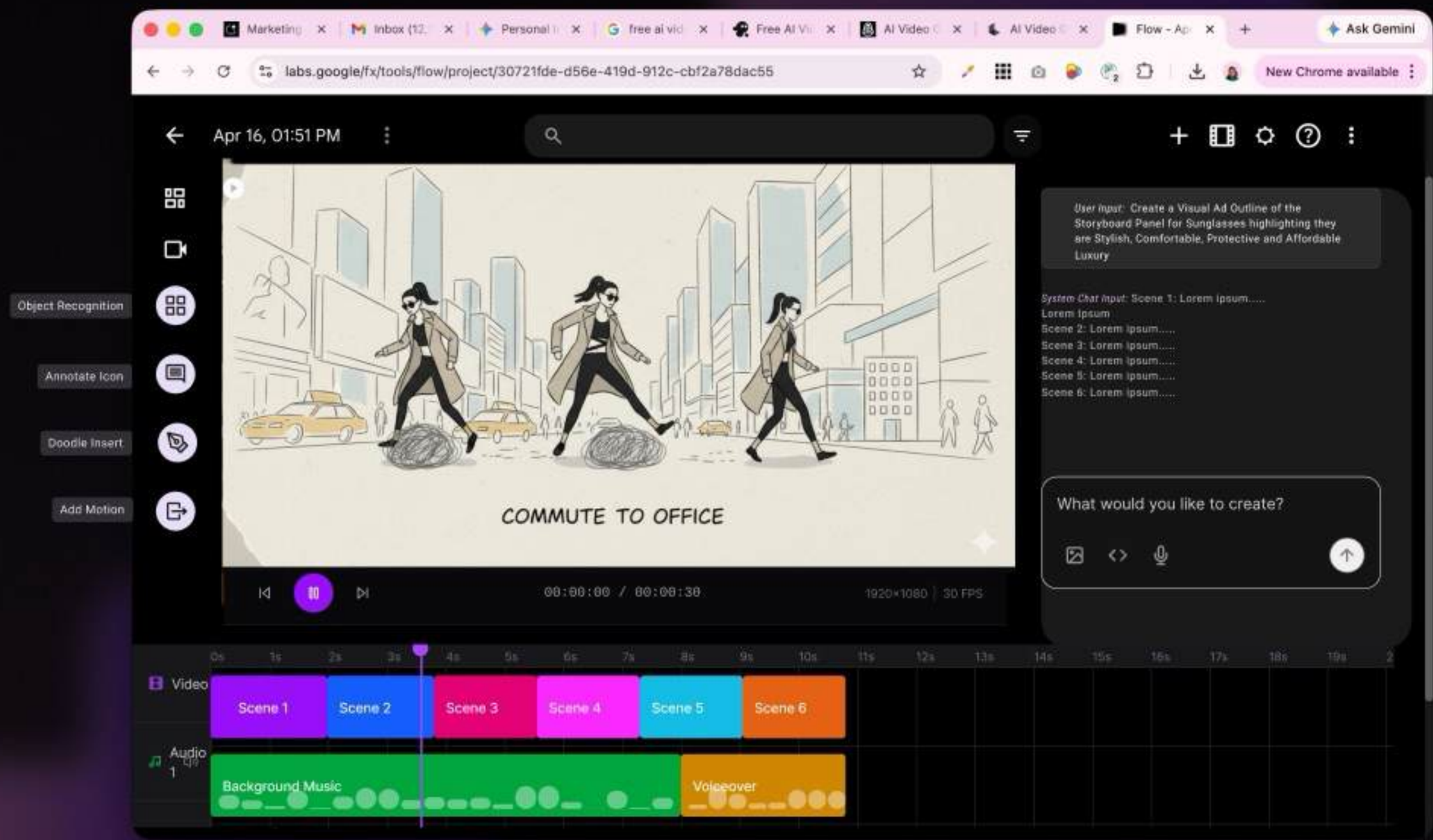


Users can **doodle to provide the finer details on design, motion in addition to chatting.** This can help the creatives express their ideas more accurately as compared to just plain text and in turn help produce better AI outcomes.

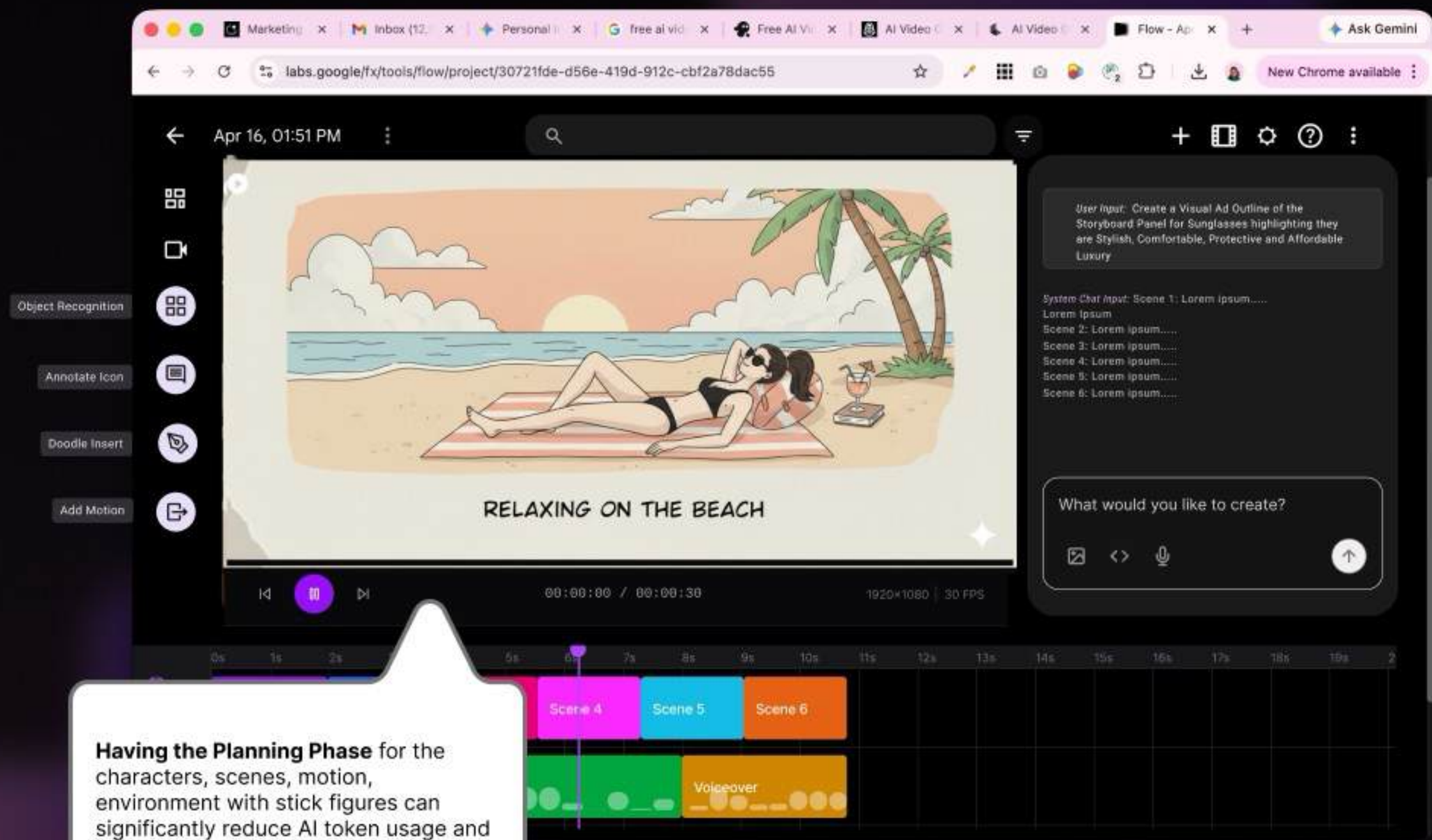
Redesign Highlights – Design for sustainable AI usage



Redesign Highlights – Design for sustainable AI usage

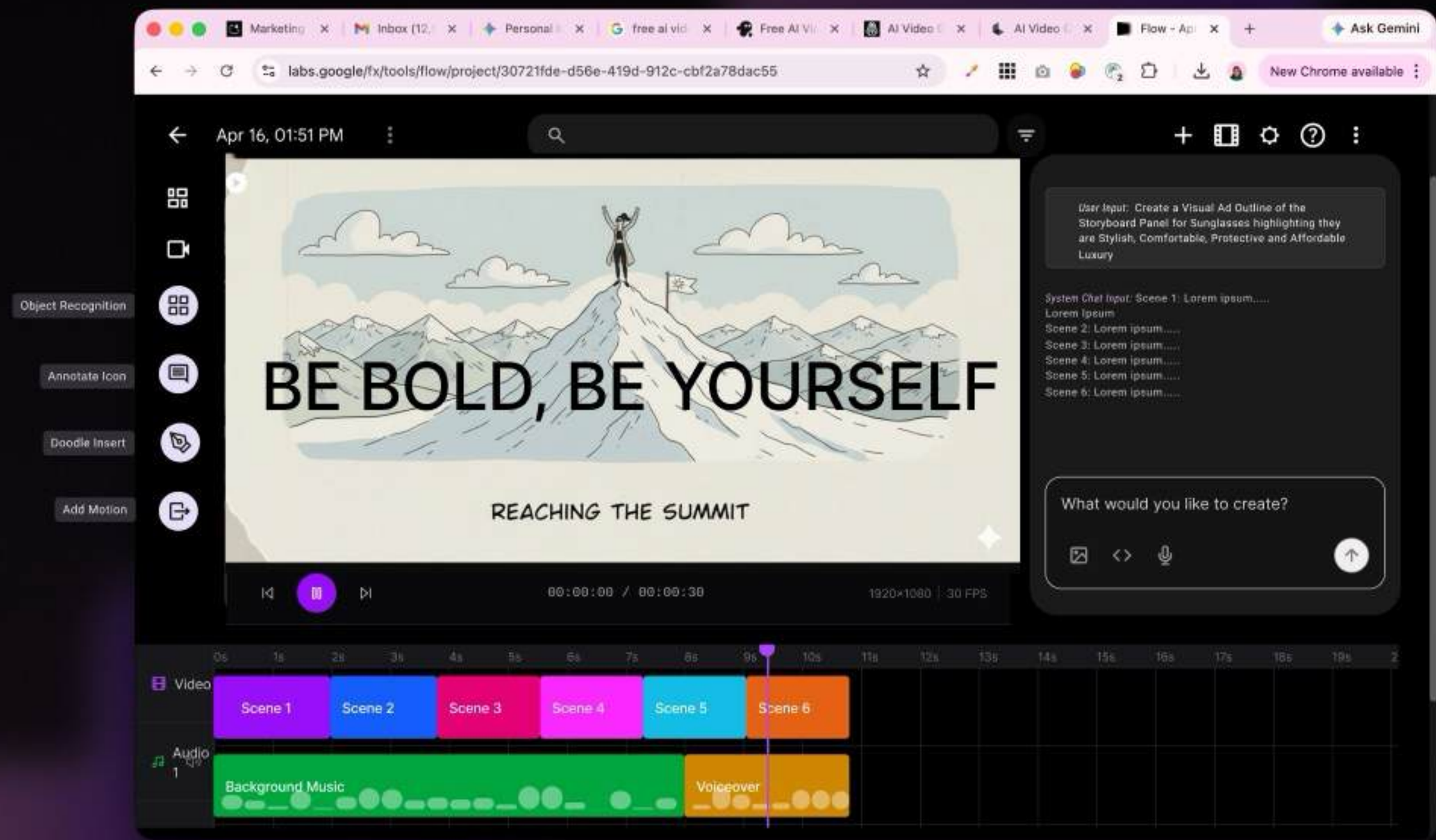


Redesign Highlights – Design for sustainable AI usage

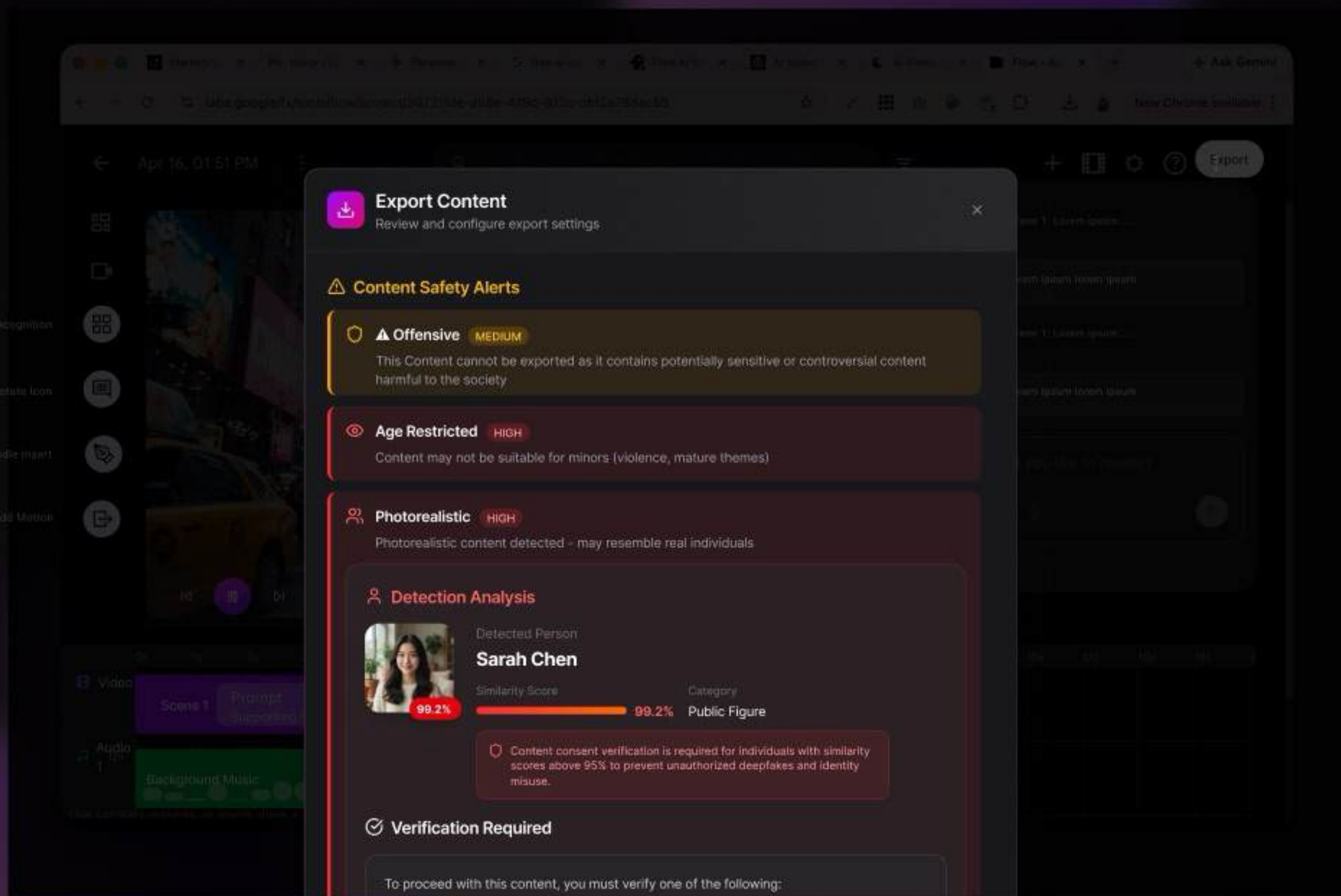


Having the Planning Phase for the characters, scenes, motion, environment with stick figures can significantly reduce AI token usage and help creatives perform low-cost iterations of their content and refine their outcome. (Low AI token usage due to NO light rendering, high resolution, 3D rendering etc.)

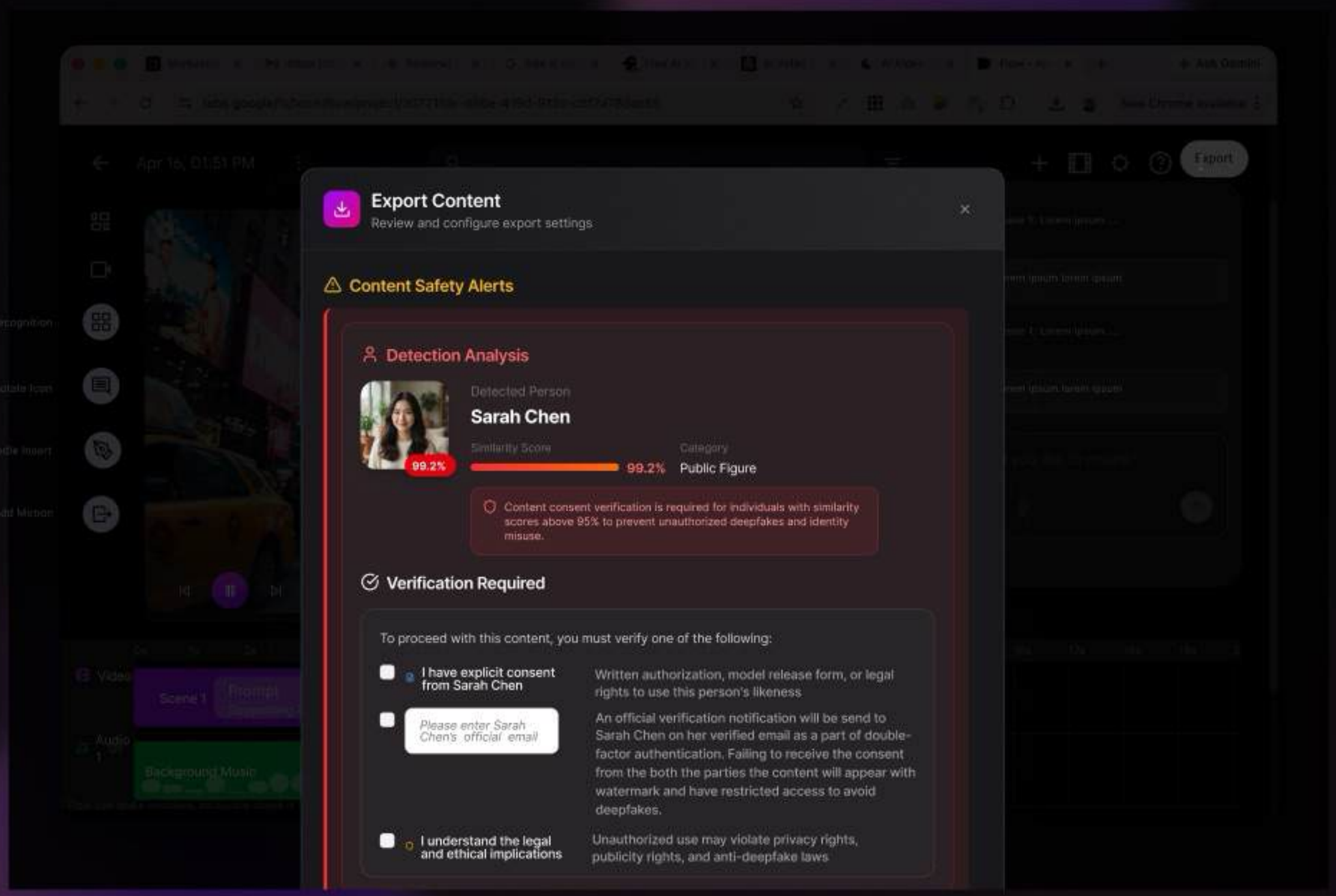
Redesign Highlights – Design for sustainable AI usage



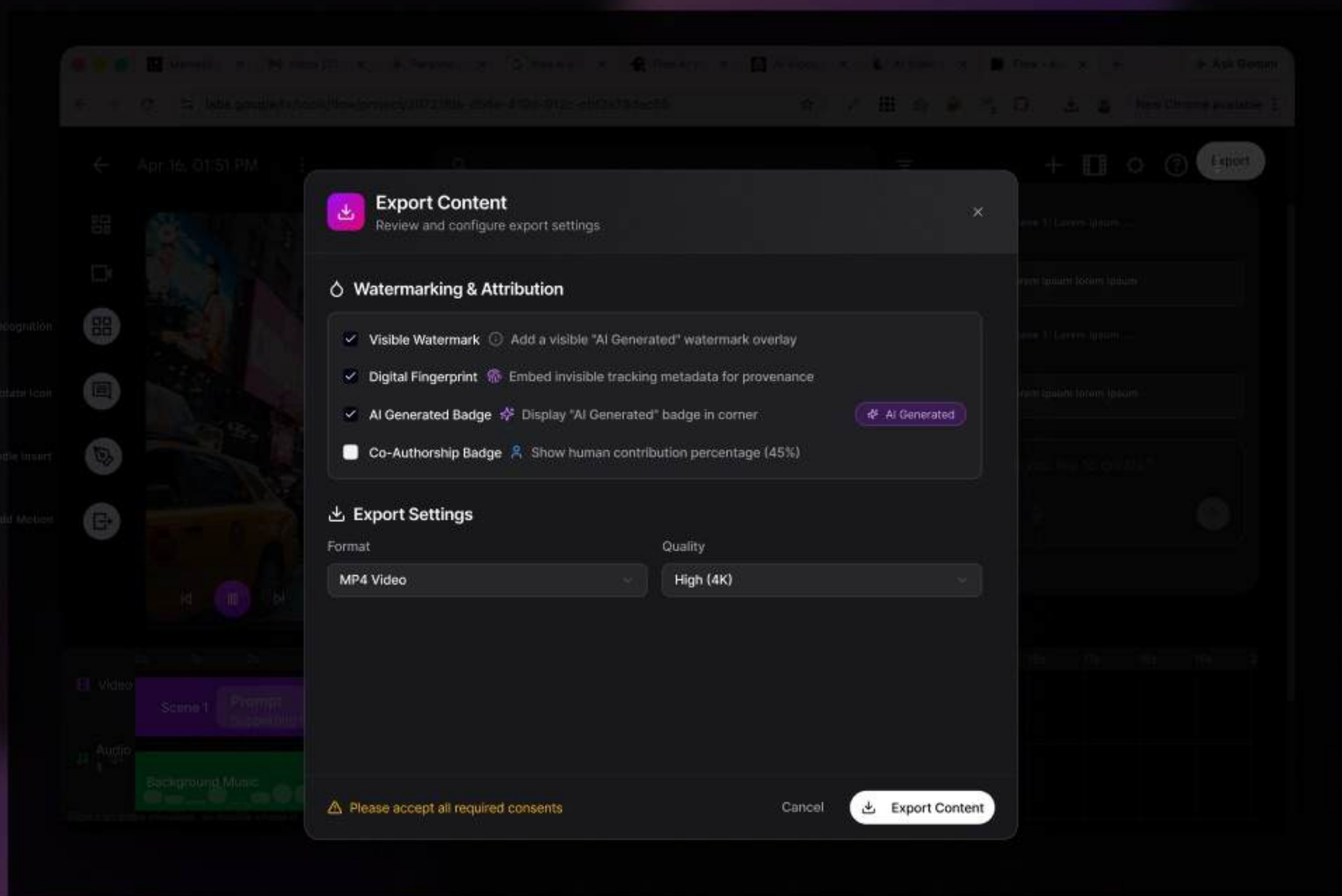
Redesign Highlights – Design to provide AI guardrails for Privacy, Safety, Security – Avoid Bad Actors, Unauthorized Content



Redesign Highlights – Design to provide AI guardrails for Security



Redesign Highlights – Design for Content Tracing & Credibility



Thank you for reading through!

I have poured a lot into envisioning a truly AI-native video editing experience with Google Flow, here.

What are your initial thoughts on this? What works for you, and what would you love to see in the creative tools of tomorrow?

Please write in the comments below, I would love to hear from you :)

Thank you !

STAY TUNED FOR MORE
SUCH CONTENT.
SUBSCRIBE TO FOLLOW

zealsheth.com